Informatica

Informatica® Cloud Application Integration
October 2024

Serialize Swift Messages with Secure Signatures

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October 2024

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Preface

Use *Serialize Swift Messages with Secure Signatures* to learn how to convert Swift XML format messages into structured Swift messages, and securely sign the messages using JSON Web Tokens (JWT).

CHAPTER 1

Introduction to Serialize Swift Messages with Secure Signatures recipe

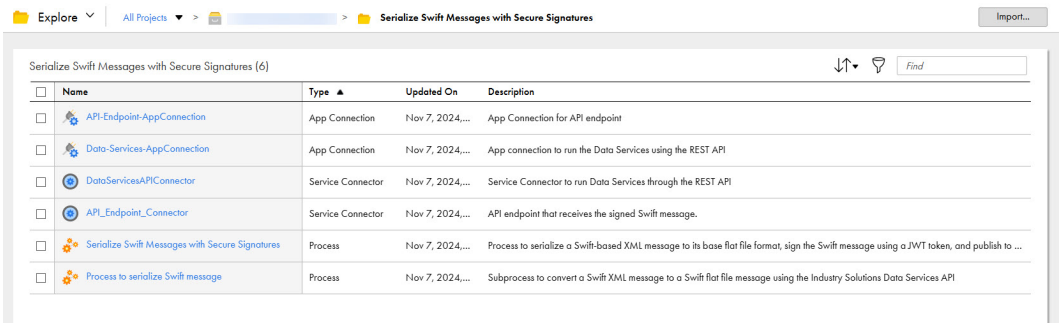
The Serialize Swift Messages with Secure Signatures recipe is based on REST and SOAP APIs. You can use the recipe to efficiently convert Swift XML format messages into structured Swift messages using the Industry Solutions Data Services API, and securely sign the messages using JSON Web Tokens (JWT).

The conversion ensures that the financial or transactional data in XML is transformed into a standard Swift message format, suitable for integration into banking or financial applications. After conversion, the recipe implements JWT signing to ensure data integrity, security, and authenticity during transmission or processing.

Serialize Swift Messages with Secure Signatures recipe contents

The Serialize Swift Messages with Secure Signatures recipe contains service connectors, app connections, and processes.

The following image shows the assets that the Serialize Swift Messages with Secure Signatures recipe package contains:



Serialize Swift Messages with Secure Signatures recipe assets

The following table lists the assets that the Serialize Swift Messages with Secure Signatures recipe package contains:

Asset Name	Asset Type	Description
API_Endpoint_Connector	Service connector	API endpoint that receives the signed Swift message.
DataServicesAPIConnector	Service connector	Service Connector to run Data Services through the REST API.
API-Endpoint-AppConnection	App connection	App Connection for the API endpoint.
Data-Services-AppConnection	App connection	App connection to run the Data Services using the REST API.
Process to serialize Swift message	Process	Subprocess to convert a Swift-based XML message to a Swift flat file message using the Industry Solutions Data Services API.
Serialize Swift Messages with Secure Signatures	Process	Process to serialize a Swift-based XML message to its base flat-file format, sign the Swift message using a JWT token, and publish to an API endpoint.

CHAPTER 2

Using the Serialize Swift Messages with Secure Signatures recipe

To use the Serialize Swift Messages with Secure Signatures recipe, you must perform the following steps manually:

1. Copy and access the recipe.
2. Publish the API_Endpoint_Connector service connector.
3. Publish the DataServicesAPIConnector service connector.
4. Configure and publish the API-Endpoint-AppConnection connection.
5. Configure and publish the Data-Services-AppConnection connection.
6. Configure and publish the processes.
7. Invoke the process.

Step 1. Copy and access the recipe

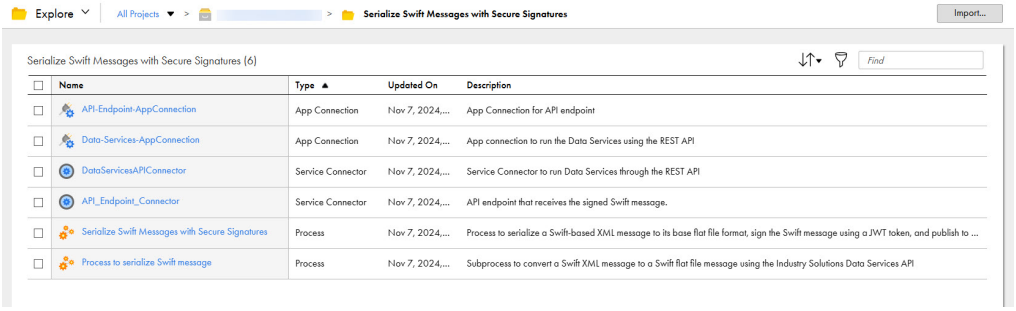
Copy the pre-configured assets in the recipe to a separate project or folder.

1. Open the **Serialize Swift Messages with Secure Signatures** recipe and click **Use**.
2. Select the location where you want to copy the recipe, and then click **Continue**.
3. In the **Copying the recipe** dialog box, click **OK**.

   It might take some time for the recipe to get copied. You will receive a notification when the recipe is ready for use.
4. After the recipe is copied, click **Explore** to access the recipe content.

5. Navigate to the project or folder where you copied the recipe or enter the recipe name in the **Find** box.

   All the assets in the recipe are displayed as shown in the following image:



Step 2. Publish the DataServicesAPIConnector service connector

Configure the service connector for the data service API to specify the URL for each action and then publish the DataServicesAPIConnector service connector.

1. Open the **DataServicesAPIConnector** service connector.
2. On the **Actions tab**, edit the **Binding** properties for each action and enter the URL.
   a. Select the **Login** action and enter the POD URL in the format `https://<POD URL>/ma/api/v2/user/login`, such as `https://dm-us.informaticacloud.com/ma/api/v2/user/login`.
   b. Select the **Invoke_Data_Service_API** action and enter the CVM URL in the format `https://<CVM URL>/DSRepo/rest/api/v1/services/run/{$serviceName}`, such as `https://usw1-cvm.dm-staging.informaticacloud/DSRepo/rest/api/v1/services/run/{$serviceName}`.
   c. Select the **Logout** action and enter the POD URL in the format `https://<POD URL>/saas/public/core/v3/logout`, such as `https://dm-us.informaticacloud.com/saas/public/core/v3/logout`.

For a complete list of POD URLs, see POD Availability and Networking on the Informatica Documentation Portal.

Save, test, and publish the service connector.

Step 3. Publish the API_Endpoint_Connector service connector

You can publish the **API_Endpoint_Connector** service connector from the assets page.

To publish the **API_Endpoint_Connector** service connector, click **Actions** in the row that contains the service connector and select **Publish**.

Step 4. Configure and publish the API-Endpoint-AppConnection connection

Configure the endpoint URL or the host name to invoke the data service API, and then publish the API-Endpoint-AppConnection connection.

1. Open the **API-Endpoint-AppConnection** connection.
2. In the **Connection Properties** section, enter the endpoint URL or the host name of the target in the **Endpoint** field.
3. Save and publish the connection.

Step 5. Configure and publish the Data-Services-AppConnection connection

Configure the user name and password to log in to Informatica Intelligent Cloud Services, and then publish the Data-Services-AppConnection connection.

1. Open the **Data-Services-AppConnection** connection.
2. From the **Runtime Environment** list, select the Secure Agent.
3. In the **Connection Properties** section, enter values for the following properties:

Property	Description
UserName	The user name to log in to Informatica Intelligent Cloud Services.
Password	The password to log in to Informatica Intelligent Cloud Services.

4. Save and publish the connection.

Step 6. Configure and publish the processes

Configure the message file details and the private key to set the Swift-based XML message file and publish the processes.

1. Open the **Process to serialize Swift message** process.
2. On the **Input Fields** tab of the **Serialize the Swift XML Message** step, enter values in the following fields:
   - Message: The message path and file name.
   - AgentGroupID: The agent group ID.
   - ErrorDir: The path to the target error directory.
   - ServiceName: The data service name to run.
3. Save and publish the process.
4. Open the **Serialize Swift Messages with Secure Signatures** process.

5. On the **Assignments** tab of the **Set the Private Key and Signing Method** step, enter the private key in the **Temp_Token_PrivKey** field. You can use any third-party tool, such as, OpenSSL, to create the private key and public key for encoding and decoding the XML messages.
6. Save and publish the process.

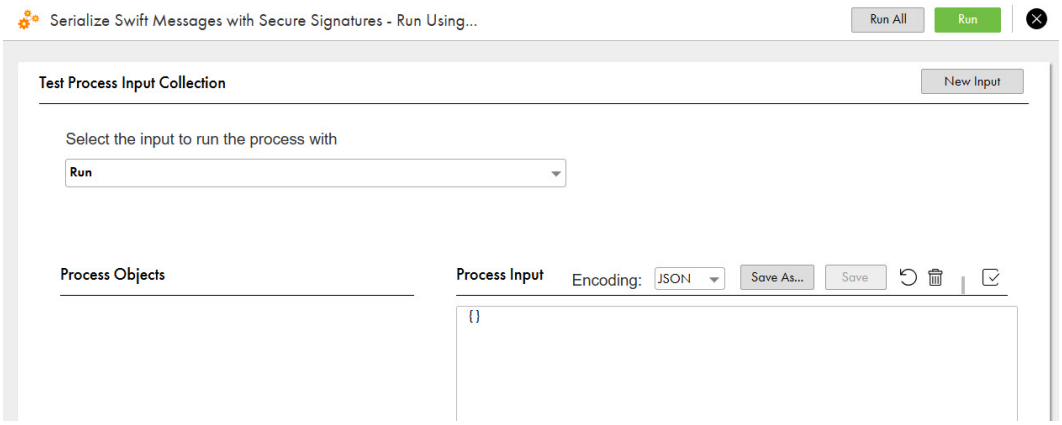
Step 7. Invoke the process

When you invoke the Serialize Swift Messages with Secure Signatures process, the process serializes a Swift-based XML message to its base flat file format, sign the Swift message using a JWT token, and publish to an API endpoint.

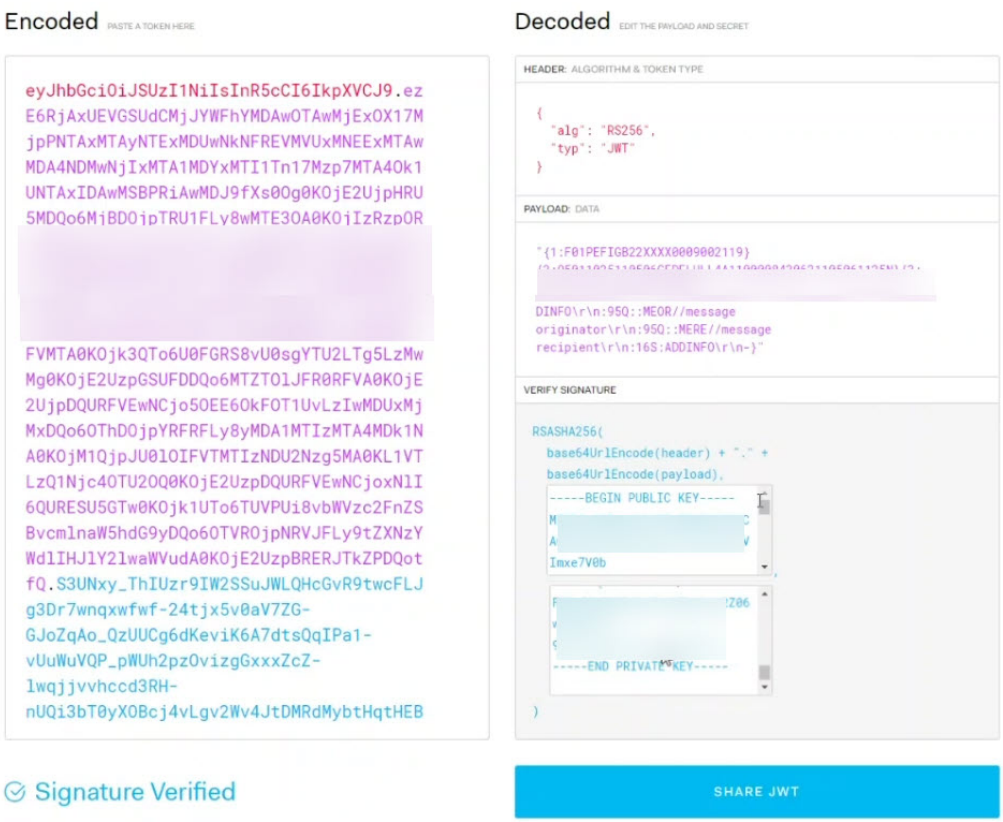
You can run the process using one of the following options:

- Run Using by passing the input parameters
- REST or SOAP API endpoints in any API client such as cURL, Postman, SOAP UI, or any programming language
- Web browser by passing the input parameters

The following image shows Run Using option to run the process:



You receive a payload in encoded format. You can use any third-party tool or JWT application to decode the payload using the private key and public key. For example, https://jwt.io/ as shown in the following image:


Encoded
PASTE A TOKEN HERE
eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9.ez
E6RjAxUEVGSUdCMjJYWFhYMDAwOTAwMjExOX17M
jpPNTAxMTAyNTExMDUwNkNFREVMVUxMNEExMTAw
MDA4NDMwNjIxMTA1MDYxMTI1Tn17Mzp7MTA4Ok1
UNTAxIDAwMSBPRiAwMDJ9fXs0Og0KOjE2UjpHRU
5MDQo6MjBDOjpTRU1FLy8wMTE3OA0KOjIzRzpOR
FVMTA0KOjk3QTo6U0FGRS8vU0sgYTU2LTg5LzMw
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A0KOjM1QjpJU0lOIFVTMTIzNDU2Nzg5MA0KL1VT
LzQ1Njc4OTU2OQ0KOjE2UzpDQURFVEwNCjoxNlI
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BvcmlnaW5hdG9yDQo6OTVROjpNRVJFLy9tZXNzY
WdlIHJlY2lwaWVudA0KOjE2UzpBRERJTkZPDQot
fQ.S3UNxy_ThIUzr9IW2SSuJWLQHcGvR9twcFLJ
g3Dr7wnqxwfwf-24tjx5v0aV7ZG-
GJoZqAo_QzUUCg6dKeviK6A7dtsQqIPa1-
vUuWuVQP_pWUh2pzOvizgGxxxZcZ-
lwqjjvvhccd3RH-
nUQi3bT0yXOBcj4vLgv2Wv4JtDMRdMybtHqtHEB
Signature Verified
Decoded
EDIT THE PAYLOAD AND SECRET
HEADER: ALGORITHM & TOKEN TYPE
{
"alg": "RS256",
"typ": "JWT"
}
PAYLOAD: DATA
"{1:F01PEFIGB22XXXX0009002119}
DINFO\r\n:95Q::MEOR//message
originator\r\n:95Q::MERE//message
recipient\r\n:16S:ADDINFO\r\n-}"
VERIFY SIGNATURE
RSASHA256(
base64UrlEncode(header) + "." +
base64UrlEncode(payload),
-----BEGIN PUBLIC KEY-----
Imxe7V0b
-----END PRIVATE KEY-----
)
SHARE JWT